ENGINEERING METROLOGY & INSTRUMENTATION

IV B. Tech. I Semester: MECHANICAL ENGINEERING								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME43	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OVERVIEW:

This course is introduced to impart knowledge of Engineering Metrology and its principle, concepts of limits, fits and tolerances. This course will focus on the basic characteristics of a typical instrument, identification of errors and their types that would occur in an instrument and the concept of transducer, various types and their character.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Identify techniques to minimize the errors in measurement.
2. Demonstrate methods and devices for measurement of length, angle, gear & amp; thread parameters, surface roughness and geometric features of parts.
3. Explain constructional details and working principles of various instruments and their purpose.
4. Demonstrate the instruments used for temperature, pressure and level.
5. Explain measurement of flow, speed, acceleration and vibration measuring instruments

UNIT-I LINEAR AND ANGULAR MEASUREMENTS

Limits, fits and tolerances- Unilateral and bilateral tolerance system, hole and shaft basis system. Interchangeability and selective assembly. Limit Gauges: Taylor's principle, Design of GO and NO-GO gauges Measurement of angles, Bevel protractor, and Sine bar. Measurements by using auto Collimator.

UNIT-II SURFACE ROUGHNESS MEASUREMENT

Roughness, Waviness. CLA, RMS, Rz Values. Methods of measurement of surface finish, Talysurf. Screw thread measurement, Gear measurement; Machine Tool Alignment Tests on lathe, Milling and Drilling machines. Coordinate Measuring Machines: Types and Applications of CMM.

UNIT-III INTRODUCTION & MEASUREMENT OF DISPLACEMENT

Introduction: Definition – Basic principles of measurement – Measurement systems, generalized configuration and functional description of measuring instruments – examples. Static and Dynamic performance characteristics – sources of errors, Classification and elimination of errors.

Measurement of Displacement: Theory and construction of various transducers to measure displacement – Piezo electric, Inductive, capacitance, resistance, ionization and Photo electric Transducers, Calibration procedures.

UNIT-IV MEASUREMENT OF TEMPERATURE , PRESSURE & LEVEL

Measurement of Temperature: Various Principles of Measurement-Classification: Expansion Type: Bimetallic Strip- Liquid in glass Thermometer; Electrical Resistance Type: Thermistor, Thermocouple, RTD.

Measurement of Pressure: Different principles used – Classification: Manometers, Dead weight pressure gauge, Tester (Piston gauge), Bourdon pressure gauges; Bellows – Diaphragm gauges. Low

pressure measurement –Thermal conductivity gauges, ionization pressure gauges, McLeod pressure gauge. **Measurement of Level:** Direct methods – Indirect methods – Capacitive, Radioactive, Ultrasonic, Magnetic level indicators – Bubbler level indicators.	
UNIT-V	**MEASUREMENT OF FLOW, SPEED , ACCELERATION & VIBRATION**
Flow measurement: Rotameter, magnetic, Ultrasonic, Turbine flow meter, **Measurement of Speed:** Mechanical Tachometers, Electrical tachometers, Non-contact type of tachometers **Measurement of Acceleration and Vibration:** Different simple instruments – Principles of Seismic instruments –Vibrometer and accelerometer using this principle – Piezo electric accelerometer.	
Text Books:	
1. M. Mahajan, A Text Book of Metrology, 1st Edition, Dhanpat Rai & Co. [P] Ltd., 2012. 2. D.S. Kumar, Mechanical Measurements and Controls, 4th Edition, Metropolitan Book Co. Pvt. Ltd. New Delhi, India, 2011. 3. K. Tayal, Instrumentation and mechanical Measurements, 2nd Edition, Galgoe Publications, New Delhi, India, 2004.	
Reference Books:	
1. Connie Dotson, Fundamentals of Dimensional Metrology, 6th Edition, Thomson Delmar, 2015. 2. Anand K Bewoor,Vinay kulakarni, Metrology & Measurement, Tata McGraw Hill, New Delhi, 2009. 3. R. K. Jain, Mechanical and Industrial Measurements, 12th Edition, Khanna Publishers, New Delhi, India, 2011. 4. Chennakesava R. Alavala, Principles of Industrial Instrumentation and Control Systems, 1st Edition, Cengage Learning, New Delhi, India, 2010. 5. B. C. Nakra, K. K. Choudhary, Instrumentation, Measurement and Analysis, 4th Edition, Tata McGraw-Hill Education Pvt. Ltd., 2016.	